# Measurement: Length, Mass & Capacity

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

**Worked examples — tutor talks through these first**

**A) Easy:** Convert 3.75kg to grams.
*Method & answer: 3.75 × 1,000 = 3,750g.*

**B) Standard:** A tank holds 18 litres. Water is used at a rate of 2.4 litres per hour. How many litres remain after 5 hours?
*Method & answer: 2.4 × 5 = 12 litres used. 18 − 12 = 6 litres remain.*

**C) Advanced:** A cable is 8.5m long. It is cut into pieces of 35cm. How many complete pieces can be cut, and how much is left over?
*Method & answer: 8.5m = 850cm. 850 ÷ 35 = 24 remainder 10, so 24 pieces with 10cm left over.*

## Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** Convert 3.75kg to grams.

**2.** Convert 12,500m to km.

**3.** Convert 0.085km to metres. A) 8.5m B) 85m C) 850m D) 8,500m

**4.** Convert 4.2l to ml.

**5.** Convert 6350g to kg.

**6.** Select ALL of these that equal 2.75 litres: A) 2,750ml B) 2l + 750ml C) 2,570ml D) 11 × 250ml E) 27.5 × 100ml

**7.** Add: 3.6m + 2.85m. Give your answer in metres.

**8.** Subtract: 8.2kg − 3.65kg.

**9.** Which of these is equal to 0.6km? A) 60m B) 600m C) 6,000m D) 0.06m

**10.** A tank holds 24 litres. Water is used at a rate of 3.5 litres per hour. How many litres remain after 4 hours?

**11.** Order from smallest to largest: 2.4kg, 2,350g, 2.45kg.

**12.** Select ALL of these masses that are heavier than 3.5kg: A) 3,450g B) 3.55kg C) 3,600g D) 3.4kg E) 3,520g

**13.** A ribbon 12.6m long is cut into pieces of 90cm each. How many complete pieces can be cut, and how much is left over?

**14.** A jug holds 1.8l. If 650ml is poured out, how much remains? A) 1.05l B) 1.15l C) 1.25l D) 1.35l

**15.** A cable is 8.5m long and is cut into pieces of 35cm each. How many complete pieces can be cut, and how much is left over?

## Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** A cyclist travels 4.65km, then a further 2,800m. What is the total distance in km?

**6.** Select ALL of these that could combine to make exactly 3 litres: A) 1.5l + 1.5l B) 1,200ml + 1,800ml C) 2l + 900ml D) 6 × 500ml E) 2,500ml + 400ml

**11.** Select ALL of these weights that round to 4.6kg to 1 decimal place: A) 4.55kg B) 4.64kg C) 4.549kg D) 4.63kg E) 4.599kg

**2.** A recipe needs 0.35kg of butter per batch. How much is needed for 6 batches, in kg?

**3.** A tank holds 45 litres and is drained at 3.75 litres per minute. How long does it take to empty? A) 10 min  B) 11 min  C) 12 min  D) 15 min

**4.** Three parcels weigh 2.4kg, 1,850g and 3.05kg. What is their total weight in kg?

**5.** A plank of wood is 4.8m long. Six shelves of 75cm each are cut from it. How much wood is left, in cm?

**7.** A gardener has 18.5m of fencing. He uses 6.2m and 5.8m for two sides. How much fencing does he have left?

**8.** A recipe for 1 loaf needs 275g of flour. A baker has 3kg of flour. How many whole loaves can be made, and how much flour is left over?

**9.** A car's fuel tank holds 52 litres. It uses fuel at 6.5 litres per 100km. How many litres are used on a 240km journey? A) 13.6l  B) 14.4l  C) 15.6l  D) 16.2l

**10.** A ribbon 15m long is cut into pieces 1.2m long. How many complete pieces can be cut, and how much is left over?

**12.** A tank contains 60 litres. It is filled at 5 litres per minute. How long does it take to fill, in minutes?

**13.** A suitcase weighs 16.8kg. Airline limit is 20kg. How much more weight can be added before the limit is reached?

**14.** A recipe scales up: 1 batch needs 0.45l of milk. How much is needed for 8 batches? A) 3.2l  B) 3.4l  C) 3.6l  D) 3.8l

**15.** A rectangular tank is filled with water at 2.4 litres per minute for 15 minutes, then drained at 3 litres per minute for 4 minutes. How much water is in the tank at the end?

## Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

**1.** A tank holds 80 litres. It is filled at 6 litres per minute for the first 8 minutes, then at 4 litres per minute after that. How long in total does it take to fill the tank?

**2.** A metal rod 2.4m long expands by 0.5% when heated. What is its new length, to the nearest mm? A) 2,410mm  B) 2,412mm  C) 2,415mm  D) 2,420mm

**3.** A recipe for 6 people needs 900g of pasta. How much pasta is needed for 15 people, keeping the ratio the same?

**4.** Select ALL of these that are within 50g of 2kg: A) 1,960g  B) 1,940g  C) 2,045g  D) 2,060g  E) 2,010g

**5.** A cyclist rides at an average speed such that they cover 3.6km every 12 minutes. How far do they travel in 50 minutes at the same

**6.** A water butt holds 150 litres. It loses water through a small leak at 0.75 litres per hour. How many litres remain after 2 full days (48 hours)?

**7.** A ribbon of length 6.3m is divided in the ratio 2:5. What is the length of the longer piece? A) 3.5m  B) 4.0m  C) 4.5m  D) 5.0m

**8.** A rectangular swimming pool has a capacity of 27,000 litres and is filled at 450 litres per minute. How many hours does it take to fill?

**9.** Three sacks of flour weigh 2.4kg, 3.75kg and 1.85kg. If the flour is repackaged equally into 4 new bags, how much does each new bag weigh, in kg?

**10.** A recipe scales in proportion to the number of people fed: 400g feeds 4 people. Select ALL of the statements below that correctly

**11.** A tank contains 45 litres of water. 30% is drained out. How many litres remain?

**12.** A plank of wood 3.6m long is cut so that one piece is twice as long as the other. Find the length of the shorter piece, in cm.

**13.** A lorry can carry a maximum load of 3.2 tonnes. It is already carrying 2,850kg. How many more 25kg bags can it carry? A) 12  B) 13  C) 14  D) 15

**14.** A recipe uses flour, sugar and butter in the ratio 5:3:2 by weight. If the total weight of these three ingredients is 1.5kg, how much flour is used, in grams?

**15.** A water tank is 60% full and contains 90 litres. What is the full capacity of the tank, in litres?

rate?

describe scaling this up: A) For 8 people, use 800g  B) For 6 people, use 500g  C) For 10 people, use 1kg  D) For 2 people, use 150g  E) For 12 people, use 1.2kg